Systematic Review

The Fundamentals and Applications of Wearable Sensor Devices in Sports Medicine: A Scoping Review

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Purpose: To (1) characterize the various forms of wearable sensor devices (WSDs) and (2) review the peer-reviewed literature of applied wearable technology within sports medicine. **Methods:** A systematic search of PubMed and EMBASE databases, from inception through 2023, was conducted to identify eligible studies using WSDs within sports medicine. Data extraction was performed of study demographics and sensor specifications. Included studies were categorized by application: athletic training, rehabilitation, and research. **Results:** In total, 43 studies met criteria for inclusion in this review. Forms of WSDs include pedometers, accelerometers, encoders (consisting of magnetometers and gyroscopes), force sensors, global positioning system trackers, and inertial measurement units. Outcome metrics include step counts; gait, limb motion, and angular positioning; foot and skin pressure; change of direction and inclination, including analysis of both body parts and athletes on a field; displacement and velocity of body segments and joints; heart rate; plethysmography; sport-specific kinematics; range of motion, symmetry, and alignment; head impact; sleep; throwing biomechanics; and kinetic and spatiotemporal running metrics. WSDs are used in athletic training to assess sport-specific biomechanics and workload with a goal of injury prevention and training optimization, as well as for rehabilitation monitoring and research such as for risk predicting and aiding diagnosis. **Conclusions:** WSDs enable real-time monitoring of human performance across a variety of implementations and settings, allowing collection of metrics otherwise not achievable. WSDs are powerful tools with multiple applications within athletic training, patient rehabilitation, and orthopaedic and sports medicine research. **Clinical Relevance:** Wearable technology may represent the missing link to quantitatively addressing return to play and previous performance. WSDs are commercially available and portable adjuncts that allow clinicians, trainers, and individual athletes to monitor biomechanical parameters, workload, and recovery status to better contextualize personalized training, injury risk, and rehabilitation.

To restore function, mobility, and performance in athletes and minimize risk of reinjury, there is an emphasis within sports medicine on injury-prevention strategies, rehabilitation, and personalized treatment. Over the past decade, there has been a push to adjunct clinical outcomes with patient-reported outcome measures (PROMs) to evaluate the efficacy of orthopaedic procedures on patient satisfaction, function, and pain.[1-3] Although becoming routine in orthopaedic clinics, the process of PROM selection, data collection, and interpretation poses challenges.[4,5] In addition, PROMs are susceptible to bias, as patient perception regarding health and physical activity varies.[6] Beyond the periodic nature and active data associated with PROMs, there is a growing need for passive and continuous data that can provide context and demonstrate patterns in patient dynamics.

Digital health care technology has significantly evolved in recent years. Wearable sensor devices (WSDs) are composed of a sensor that produces electrical signal in response to a physical factor, along with a transducer (an electronic circuit) that analyzes and transmits this signal wirelessly.[7] Through integration into accessories and clothing, or worn directly, WSDs allow for objective, continuous evaluation of physical activity and health outcomes.[8] Today, WSDs range from sports and fitness trackers to Food and Drug Administration–regulated medical devices.[9,10] Examples include the Fitbit (San Francisco, CA), Apple Watch (Cupertino, CA), CricFlex elbow sleeve

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(Islamabad, Pakistan), and Polar chest strap (Kempele, North Ostrobothnia, Finland).[10] In 2018, WSD sales volume reached 27.9 million units and are expected to continue to rise by 13.8% before 2028.[11-13] This explosive growth is largely attributed to technological advancements. Electromechanical sensors convert human movements into kinematic and kinetic parameters. Traditional sensors collecting this information include pedometers, accelerometers, encoders (gyroscopes, magnetometers), and force sensors—all of which can assess training load.[14] These may be combined with global positioning system (GPS) technology, which detects limb motion, to form even more sophisticated sensors such as inertial measurement units (IMUs) that enable real-time motion detection, indoor navigation, and activity recognition.

The advancements in WSDs, analytics, and durability have positioned this sector to be implemented in clinical context. However, there is limited research on WSDs in patient recovery and trajectory and poor consensus on usage guidelines. Furthermore, despite the wide array of wearables, including wrist monitors, sensors have been typically applied near the patient's center of mass for activity analysis.[15] Although challenging to incorporate novel technologies into practice, clinicians perceive significant value in the use of WSDs.[16] The purpose of this review is to (1) characterize the various forms of WSDs and (2) review the peer-reviewed literature of applied wearable technology within sports medicine. This can offer clinicians a contemporary, educational resource to inform decision-making regarding WSD integration.

Methods

Due to the exploratory and broad nature of our research focus, a scoping review was conducted. A systematic approach was guided by Arksey and O'Malley's[17] methodological framework for scoping reviews and recommendations by the Joanna Briggs Institute.[18] Study selection was reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews checklist.[19] Formal methodological quality and risk of bias assessments were not performed, as scoping reviews do not intend to critically appraise a cumulative body of evidence but rather provide an overview of existing evidence.[19,20]

Studies were included if they were original, peer-reviewed research articles reporting the use and application of WSDs within sectors of sports medicine. Studies reporting on sensor usage in arthritic populations or following arthroplasty were excluded. Explicit protocol and technology validation studies, as well as all review articles, also were excluded.

In December 2023, an electronic database literature search was conducted using PubMed and EMBASE, from inception to present. The following search terms were used: ("wearable device" OR "wearable technology" OR "wearable sensor") AND ("sports medicine"). In addition, a manual search was performed of reference lists from relevant systematic reviews. Search results were exported to a reference management software for removal of duplicates. Title and abstract screening was conducted by 2 independent raters (R.J.O. and S.S.H.). Remaining full-texts were evaluated for inclusion in the final review. A third reviewer (P.N.R.) was available for consultation.

Data were extracted for all included studies, consisting of study demographics, technology specifications, sensor placement, and sensor-derived outcome variables. For studies with injured or postoperative cohorts, follow-up time from injury or surgical intervention was recorded. To facilitate charting and focus our results, studies were categorized into 1 of 3 subgroups based on their application of WSDs: athletic training, rehabilitation, and future research directions.

Results

Identification of Studies

A Preferred Reporting Items for Systematic Reviews flow diagram[21] detailing the study selection process is presented in Figure 1. After removal of duplicates, our database search returned 742 records for title and abstract screening. Seven records were manually added from reference lists of systematic reviews. A total of 147 full-texts were screened, of which 43 studies remained for inclusion in this review.

Characterization of WSDs

Pedometers

Pedometers are the most basic and prevalent type of motion sensor. When the vertical acceleration of the lever arm exceeds the pedometer's force sensitivity threshold, a "step" is documented. Pedometers have been historically effective in quantifying step counts and improving fitness levels among the general population but have limited applications in sports medicine and biomechanics due to their inability to monitor changes in direction or energy expenditure.[22]

Accelerometers

Accelerometers quantify the acceleration, or rate of velocity change, of an object and provide data on static and dynamic forces acting on the object. They consist of 2 elements: a mechanical motion-detecting device and a microchip that evaluates signals from the mechanical device. Primary types of accelerometers include piezoresistive, piezoelectric, and differential capacitors. Piezoresistive and piezoelectric accelerometers detect changes in the force of acceleration as resistance and electrical charges, respectively, proportional to applied

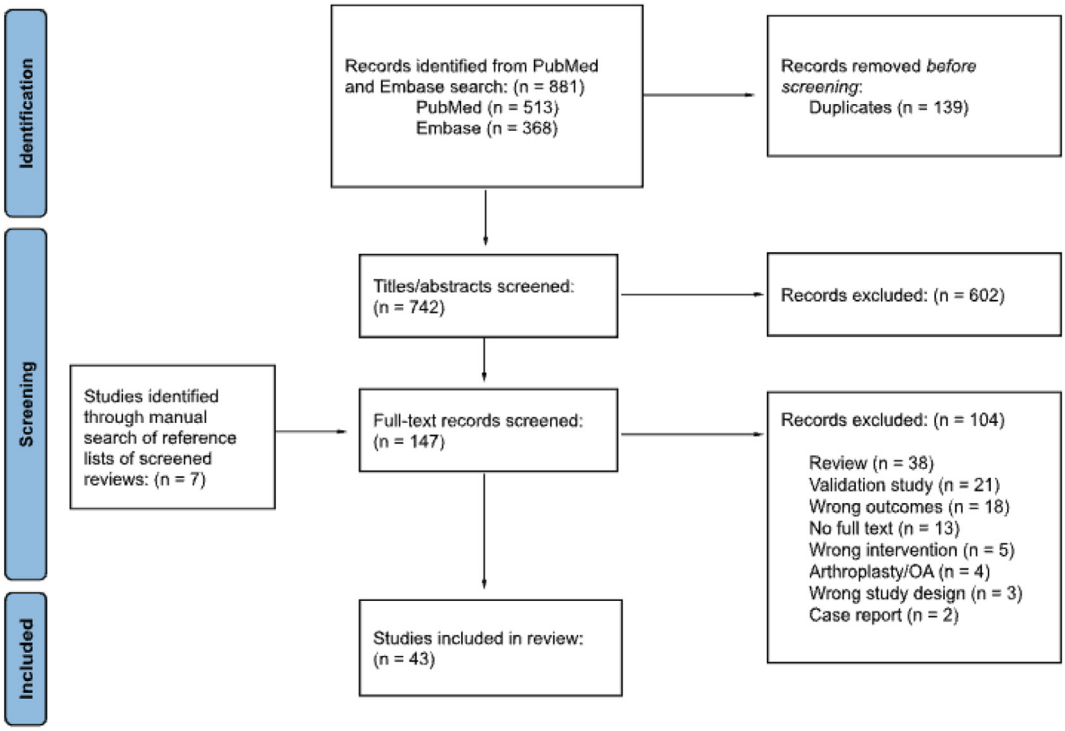

Fig 1. PRISMA literature review flow diagram.[21] (PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.)

mechanical stress. Differential capacitors analyze changes in capacitance proportional to acceleration. The accelerometer consists of 2 sets of parallel plates, with one set fixed and the other attached to a movable mass, which is directly influenced by acceleration. As the acceleration changes, the separation between the fixed and movable plates changes, resulting in a change in capacitance. Differential capacitors are commonly used in sports due to their higher sensitivity and improved noise rejection compared to piezoresistive and piezoelectric accelerometers.[23] Accelerometers have been used extensively in gait analysis, limb motion, and sports studies.[24-32]

Advances in microelectromechanical systems (MEMS) devices have enabled the integration of multiple transducers, which allows a single sensor to detect and interpret motion in various dimensions.[33] Triaxial accelerometers can measure acceleration using piezoelectric techniques that are processed to determine acceleration along each axis.[34] Triaxial accelerometers can also detect energy expenditure by integrating acceleration over time.[35]

Encoders

Encoders consist of magnetoresistive sensors (magnetometers) and gyroscopes. Magnetometers use the direction and strength of magnetic fields to provide information about body movement, analyzing variances in the resistivity of a current, with the changes proportional to the tilt angle in relation to the magnetic field direction.[36,37] Gyroscopes track orientation changes, resulting in angle and angular velocity measurements that can be calculated by integrating the linear motion detected with the angular rate of rotation in the reference frame. This value can be used to ascertain object positioning.[38] Such devices have previously been used to obtain angular information on various lower-limb joints.[39] Like accelerometers, these can be packaged as MEMS devices, resulting in triaxial magnetometers and gyroscopes.

Force Sensors

Force sensors measure forces applied by the subject. The 2 primary types are load cells and force-sensitive resistors (FSRs). Load sensors consist of a metal structure with strain gauges attached. When force is applied to the load cell, the metal structure deforms, causing the strain gauges to change resistance. This change in resistance is converted into an electrical signal that represents the applied force.[40] FSRs are piezoresistive force sensors that measure changes in the electrical resistance of a material in response to applied stress.[41] FSRs are highly sensitive, low-cost, thin, and flexible, making them suitable for integration into electronic circuits and allowing for a more comfortable and less obtrusive WSD design.

Ground reaction force is a measure of the force between the subject and the ground. Ground reaction

force plays a crucial role in human movement, as the interaction between the ground and subject determines balance, energy transfer, force generation, and distribution along the muscles and joints—which can subsequently determine joint torque.[42] FSRs are commonly placed within the insole of the individual's footwear, as pressure insoles, enabling precise evaluation of the 3-dimensional characteristics of the plantar foot surface.[43,44] Similarly, in upper limbs, muscle contractions and subsequent distortions in the skin exert pressure on the FSR to be converted into force.[45]

GPS Trackers

Through a synchronized transfer of signals between the wearable GPS receiver and orbiting satellites, this technology can quantify comparative velocity and precise location. This allows for the analysis of movement in all dimensions, including specific changes in direction and inclination.[46] GPS wearables have been extensively used in measuring positional data in athletics.[47,48] To date, however, the majority of research using WSDs in team sports has been conducted outdoors due to the inability of GPS units to track movement patterns indoors.[49]

Inertial Measurement Unit (IMU)

IMUs may effectively and accurately analyze gait in a timely and cost-efficient manner.[50,51] IMUs are MEMS-based devices composed of triaxial accelerometers, gyroscopes, and occasionally magnetometers to provide a comprehensive measure of external training load.[52] These devices integrate elements of conventional movement analysis methods into portable wearables and are energy-efficient, compact, lightweight, and economical.[53] IMUs serve to (1) record continuous physical activity data in body motion analysis and (2) measure the displacement and velocity of body segments and joints.[14]

Because IMUs are unable to measure absolute positioning over long periods of time, many are combined with GPS devices to facilitate more precise locomotion measures.[54,55] Another constraint is their inability to estimate the weight of a lifted object, restricting their capacity to identify injury risk resulting from physical overload or excessive exertion. IMUs therefore can be combined with pressure insoles for improved accuracy.

Sensor Types and Use

Complete data regarding sensor models and specifications, body placement, and sensor-derived outcome variables for all included studies are detailed in Table 1.[55-98] The most used WSD was an IMU consisting of a triaxial accelerometer and triaxial gyroscope (n = 31), of which 6 also included a triaxial magnetometer. One study used a single-axis accelerometer.[56] Other WSDs included heart rate monitors, GPS, pedometers, and photoplethysmography. Three studies used plantar pressure insoles.[57-59] Sampling frequencies were customizable and variable, ranging from 15 Hz to 3,200 Hz. One study collected activity data from pre-existing smartphone or body-worn sensors already used by patients before their injury.[60]

Most studies used just a single WSD, and sensors were attached to a variety of anatomical locations (Table 1). Two studies used a wireless motion capture suit (Xsens MVN) embedded with 17 IMUs.[61,62] Plantar pressure insoles and shoelace-mounted IMUs were worn bilaterally. Most of these devices were applied on-field (track, court, pool). Other settings included laboratories, athletic training rooms or clinics, and patients' homes. Two studies used a smartphone application in conjunction with WSDs for patient education of WSD use and automatic calculations from sensor-derived data.[63,64] These applications (Orthelligent HOME[63], iShould[64]) included test and exercise instructional videos and evaluated performance.

Sensor-derived outcome metrics included spatiotemporal gait parameters, sport-specific kinematics, step count, range of motion, symmetry and alignment, head acceleration magnitudes, sleep, heart rate, and plantar pressure (Table 1). Five studies analyzing throwing biomechanics used a Motus Global compression sleeve worn on the throwing arm to measure arm slot, arm speed, shoulder rotation, and elbow torque.[65-69] RunScribe Plus sensors (RunScribe Labs), mounted to the heel or shoelaces, were a common intervention to measure kinetic and spatiotemporal running metrics such as pace, cadence, ground contact time, stride length, pronation excursion, impact and braking acceleration, and foot strike type.[70-73]

To address growing concussion concerns, head impact was a primary outcome variable in 10 studies.[56,74-82] Six of these used the X-patch (X2 Biosystems), placed directly on the skin over the mastoid process, to measure linear and rotational acceleration.[75,76,78,80-82] Other studies embedded sensors within players' helmets or mouthguards.[56,74,77,79] Only 2 studies did not cross-reference video recording for verification of head impacts.[56,80]

Athletic Training

Most studies included in this review (n = 20) used WSDs for athletic training purposes, to assess sport-specific biomechanics, workload, and injury-prevention strategies. All studies evaluated healthy athletes, with one exception.[71] Excluding one laboratory-based study,[57] all technology was applied on-field (track, court, pool). Study demographics are found in Table 2.[56,57,66,69-78,80,81,83-87]

Table 1. Sensor Specifications and Utility

Study	Sensor Type	Model	Sampling Frequency	No. of Sensors	Body Location	Setting	Sensor-Derived Outcome Measures	Application Category
Benson et al., 2021[85]	Triaxial accelerometer, triaxial gyroscope, triaxial magnetometer	VERT (Mayfonk Inc., Fort Lauderdale, FL)	Not recorded	1	COM	On-field	Jump count, jump height, weighted jump height, session rating of perceived exertion	Athletic training
Brooks et al., 2021[74]	Triaxial accelerometer, triaxial gyroscope	GForce Tracker (Artaflex Inc., Markham, ON, Canada)	Not recorded	1	Helmet	On-field	Head impact (linear and rotational acceleration)	Athletic training
Carey et al., 2021[75]	Triaxial accelerometer, triaxial gyroscope	X-patch (X2 Biosystems Inc., Seattle, WA)	Not recorded	1	Mastoid process	On-field	Head impact (linear and rotational acceleration)	Athletic training
Caswell et al., 2017[78]	Triaxial accelerometer, triaxial gyroscope	X-patch (X2 Biosystems Inc., Seattle, WA)	Not recorded	1	Mastoid process	On-field	Head impact (linear and rotational acceleration)	Athletic training
Caswell et al., 2019[77]	Triaxial accelerometer, triaxial gyroscope	GForce Tracker (Artaflex Inc., Markham, Ontario, Canada)	3,000 Hz (linear acceleration), 760 Hz (rotational velocity)	1	Helmet	On-field	Head and body impacts (linear and rotational acceleration)	Athletic training
Caswell et al., 2020[76]	Triaxial accelerometer, triaxial gyroscope	X-patch (X2 Biosystems Inc., Seattle, WA)	Not recorded	1	Mastoid process	On-field	Head impact (linear and rotational acceleration)	Athletic training
DeJong Lempke et al., 2021[71]	Triaxial accelerometer, triaxial gyroscope	RunScribe Plus (RunScribe Labs, Half Moon Bay, CA)	200 Hz	2	Heel	On-field	Stride length, contact time, cadence, flight ratio, shock, impact, braking, pronation excursion, maximum pronation velocity	Athletic training
DeJong Lempke et al., 2022[70]	Triaxial accelerometer, triaxial gyroscope	RunScribe Plus (RunScribe Labs, Half Moon Bay, CA)	200 Hz	2	Shoelaces	On-field	Pace, cadence, contact time, stride length, maximum pronation velocity, shock, foot strike type, impact g, braking g, pronation excursion	Athletic training
Freehill et al., 2023[84]	Triaxial accelerometer, triaxial gyroscope, triaxial magnetometer	GT9X (ActiGraph, LLC, Pensacola, FL)	100 Hz	1	Midhumerus, 3 to 5 cm above the lateral epicondyle	On-field	Throw/pitch count and intensity	Athletic training
Fujishita et al., 2023[57]	In-shoe plantar pressure system, triaxial accelerometer, triaxial gyroscope	Pedar-X system (Novel, Munich, Germany); Physilog5 (Gait Up Ltd., Lausanne, Switzerland)	100 Hz (plantar pressure system); 128 Hz (IMU)	4 (*99 capacitive sensors in Pedar system)	Insole, shoelaces, waist	Laboratory	Plantar pressure distribution, gait cycle time, cadence, stance, swing ratios, stride length, speed, foot pitch angle, peak angular velocity	Athletic training

(continued)

Table 1. Continued

Study	Sensor Type	Model	Sampling Frequency	No. of Sensors	Body Location	Setting	Sensor-Derived Outcome Measures	Application Category
Khalil et al., 2021[66]	Triaxial accelerometer, triaxial gyroscope	motusTHROW (Motus Global, Seattle, WA)	1000 Hz	1	1.5 in distal to medial epicondyle of humerus	On-field	Arm slot, arm speed, shoulder rotation, elbow varus torque	Athletic Training
King et al., 2018[80]	Triaxial accelerometer, triaxial gyroscope	X-patch (X2 Biosystems Inc., Seattle, WA)	1,000 Hz (linear acceleration), 800 Hz (angular velocity)	1	Mastoid process	On-field	Head impact (linear and rotational acceleration)	Athletic training
Lauck et al., 2022[56]	Single-axis accelerometer, GPS	HIT System (Riddell, Elyria, OH); Catapult Vector GPS (Catapult Sports, Melbourne, Australia)	1,000 Hz (accelerometer array), 10 Hz or 18 Hz (LPS or GPS)	2 (*accelerometer array contains 6 single-axis accelerometers)	Helmet; halter vest	On-field	Head impact events (linear and rotational acceleration), repetitive head impact exposure, total player load, load rate, session time	Athletic training
Li et al., 2019[87]	Triaxial accelerometer, GPS	Catapult OptimEye (Catapult Sports, Melbourne, Australia)	15 Hz (GPS), 100 Hz (accelerometer)	2	In harness between scapulae	On-field	Physical workload calculated from acceleration	Athletic training
Long et al., 2023[73]	Triaxial accelerometer, triaxial gyroscope	RunScribe Plus (RunScribe Labs, Half Moon Bay, CA)	200 Hz	2	Shoelaces	On-field	Impact, stride length, braking, total shock, cadence, ground contact time	Athletic training
Mehta, 2019[69]	Triaxial accelerometer, triaxial gyroscope	motusTHROW (Motus Global, Seattle, WA)	1,000 Hz	1	1.5 in distal to medial epicondyle of humerus	On-field	Arm slot, arm speed, shoulder rotation, elbow varus torque	Athletic training
Pexa et al., 2023[86]	Triaxial accelerometer, GPS, heart rate monitor	Team Polar Pro (Polar Electro Oy, Kempele, Finland)	Not recorded	1	Beneath xiphoid process	On-field	Distance, speed, sprint, heart rate, accelerations/ decelerations, training impulse, calories	Athletic training
Hamidi Rad et al., 2021[83]	Triaxial accelerometer, triaxial gyroscope	Physilog IV (GaitUp, CH); SpeedRT (ApLab, Rome, Italy)	500 Hz	1	Sacrum	Swimming pool	Displacement, velocity, and duration in each swim phase, stroke rate, # of kicks	Athletic training
Reyes et al., 2020[81]	Triaxial accelerometer, triaxial gyroscope	X-patch (X2 Biosystems Inc., Seattle, WA)	1,000 Hz	1	Mastoid process	On-field	Head impact (linear and rotational acceleration)	Athletic training
Willmott et al., 2018[82]	Triaxial accelerometer, triaxial gyroscope	X-patch (X2 Biosystems Inc., Seattle, WA)	1,000 Hz	1	Mastoid process	On-field	Head impact (linear and rotational acceleration)	Athletic training
Aufwerber et al., 2019[59]	Plantar force sensor insole, pedometer	Loadsol Insoles (Novel Inc., Pittsburgh, A); Yamax SW 200/LS2000 (Yamax Corporation, Tokyo, Japan)	Not recorded	3	Hip; insole	Patient home	Step count, plantar force loading magnitude and frequency	Rehabilitation

(continued)

Table 1. Continued

Study	Sensor Type	Model	Sampling Frequency	No. of Sensors	Body Location	Setting	Sensor-Derived Outcome Measures	Application Category
Bailey et al., 2016[94]	triaxial accelerometer	3-Space Wireless Sensor (YEI Technology, Portsmouth, OH)	200 Hz	1	Lumbar spine L3	Clinic	Acceleration profiles from step-up-and-over test	Rehabilitation
Bell et al., 2017[92]	Triaxial accelerometer	wGT3X-BT (ActiGraph, LLC, Pensacola, FL)	Not recorded	1	Anterior superior iliac spine	Patient home	Step count, time spent in moderate-to-vigorous physical activity (MVPA)	Rehabilitation
Dan et al., 2019[97]	Triaxial accelerometer, triaxial gyroscope	ViMove (DorsaVi, East Melbourne, Australia); ST Microelectronics LSM303DLHC (Geneva, Switzerland)	100 Hz (x-axis), 20 Hz (y- and z-axis)	2	Pretibial border	Laboratory	Flight time, varus/valgus displacement, tibial inclination range, speed of movement	Rehabilitation
DeJong Lempke et al., 2022[72]	Triaxial accelerometer, triaxial gyroscope, triaxial magnetometer	RunScribe Plus (RunScribe Labs, Half Moon Bay, CA)	200 Hz	2	Shoelaces	Laboratory, on-field	Contact time, cadence, pace, stride length, shock, pronation excursion and velocity	Rehabilitation
Dowling et al., 2018[65]	Triaxial accelerometer, triaxial gyroscope	motusBASEBALL (Motus Global, Seattle, WA)	1,000 Hz	1	Medial ulna 5 cm below medial epicondyle of humerus	Laboratory	Arm slot, arm speed, shoulder rotation, elbow varus torque	Rehabilitation
Ezzat et al., 2021[93]	Triaxial accelerometer	GT3X (ActiGraph, LLC, Pensacola, FL)	Not recorded	1	Anterior superior iliac spine	Patient home	Time spent in physical activity (PA) levels: MVPA, vigorous, moderate, light	Rehabilitation
Fino et al., 2019[89]	Triaxial accelerometer, triaxial gyroscope	Opal v1 (APDM, Inc. Portland, OR)	128 Hz	4	Forehead, lumbar spine, anterior distal region of each shank	Athletic training room	Head angular velocity, head angle (ROM), gait speed	Rehabilitation
Gadangi et al., 2023[91]	Triaxial accelerometer, photoplethysmography	WHOOP 2.0 Strap (Boston, MA)	Not recorded	1	Wrist	Patient home	Heart rate, heart rate variability, ambient temperature, motion and movement, skin response, sleep stage, duration asleep/awake	Rehabilitation
Leafblad et al., 2019[67]	Triaxial accelerometer, triaxial gyroscope	motusBASEBALL (Motus Global, Seattle, WA)	1,000 Hz	1	5 cm distal to medial epicondyle of humerus	Laboratory	Arm slot, arm speed, shoulder rotation, elbow varus torque	Rehabilitation
Lizzio et al., 2021[68]	Triaxial accelerometer, triaxial gyroscope	motusTHROW (Motus Global, Seattle, WA)	1,000 Hz	1	1.5 in distal to medial epicondyle of humerus	Laboratory	Arm slot, arm speed, shoulder rotation, medial elbow torque	Rehabilitation

(continued)

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Table 1. Continued

Study	Sensor Type	Model	Sampling Frequency	No. of Sensors	Body Location	Setting	Sensor-Derived Outcome Measures	Application Category
Mengis et al., 2023[63]	Triaxial accelerometer, triaxial gyroscope	Orthelligent (OPED GmbH, Valley, Germany)	100 Hz	1	Tibial tuberosity	Patient home	ROM, knee displacement, balance, jump height, jump distance, no. of hops, time of test completion	Rehabilitation
Moretti et al., 2023[96]	Triaxial accelerometer, triaxial gyroscope	Not recorded	300 Hz	1	Back (level of the hip)	Laboratory	Balance, strength, agility, velocity, symmetry, fatigue	Rehabilitation
Neely et al., 2022[90]	Triaxial accelerometer, photoplethysmography	Fitbit Charge 3 (FitBit, San Francisco, CA)	Not recorded	1	Wrist	Patient home	Sleep time, wake time, time spent awake in bed at night, average steps/day, exercise frequency, exercise duration	Rehabilitation
Setuain et al., 2015[95]	Triaxial accelerometer, triaxial gyroscope	MTx, 3DOF Human Orientation Tracker (Xsens Technologies B.V., Enschede, Netherlands)	100 Hz	1	Lumbar spine L3	On-field	Peak acceleration and angular excursion from vertical jump test battery	Rehabilitation
Trbovich et al., 2021[88]	Triaxial accelerometer	GT3x+ (ActiGraph, LLC, Pensacola, FL)	Not recorded	1	Wrist	Patient home	Total sleep time, sleep efficiency	Rehabilitation
Braun et al., 2023[60]	*variable (patient's pre-existing smartphone or body-worn sensor)	Apple device (Cupertino, CA) or Google operating system (Mountain View, CA)	Not recorded	1	Not recorded	Patient home	Step count	Research
Brund et al., 2021[62]	Triaxial accelerometer, triaxial gyroscope, triaxial magnetometer, heart rate monitor, GPS	Xsens MVN Link (Xsens Technologies B.V, Enschede, The Netherlands); Garmin Forerunner 735XT; HRM-Run (Garmin Ltd., Olathe, KS)	240 Hz (motion capture); 60 Hz (Garmin)	18	Head, sternum, pelvis, upper legs, lower legs, feet, shoulders, upper arms, forearms, hands, wrist	On-field	Cadence, speed, intercept, vertical oscillation, ground contact time, stride length, cadence, heart rate	Research
Di Paolo et al., 2023[61]	Triaxial accelerometer, triaxial gyroscope, triaxial magnetometer	MVN Lycra (Xsens Technologies, Enschede, Netherlands)	240 Hz	17	Head, sternum, hands, wrists, elbows, middle of scapula spines, pelvis, thighs, tibias, forefoot	Laboratory, on-field	Hip, knee, ankle, and pelvis joint kinematics, ROM, and alignment, foot contact, cut speed	Research

(continued)

Table 1. Continued

Study	Sensor Type	Model	Sampling Frequency	No. of Sensors	Body Location	Setting	Sensor-Derived Outcome Measures	Application Category
Elstub et al., 2023[58]	Plantar pressure insole, triaxial accelerometer, triaxial gyroscope, triaxial magnetometer	Pedar insoles (Novel, Pittsburgh, PA); Xsens MVN Awinda (Xsens Technologies B.V, Enschede, The Netherlands)	100 Hz	9	Insole, shoelaces, pelvis, thighs, shanks	Laboratory	Plantar force, acceleration, angular velocity, peak tibial compression force	Research
Goodin et al., 2021[79]	Triaxial accelerometer (3x array), triaxial gyroscope	Nexus A9 (HitIQ Pty. Ltd., Canberra, Australia); ADXL372 accelerometers (Analog Devices, Inc., Wilmington, MA); Bosch BMG 250 gyroscope (Bosch Sensortec, Reutlingen, Baden-Wurttemberg, Germany)	3200 Hz (accelerometers), 800 Hz (gyroscope), *all resampled to 3,200 Hz	4	Mouth	On-field	Head and body impacts (linear and rotational acceleration)	Research
McGeown et al., 2022[98]	Triaxial accelerometer, three-lead ECG	Zephyr BioHarness (Medtronic, Dublin, Ireland)	250 Hz (ECG), 100 Hz (accelerometer *upsampled to 250 Hz for analysis)	1	Level of xiphoid process	Clinic	Heart rate, accelerometry time series data	Research
Pichonnaz et al., 2015[64]	Triaxial accelerometer, triaxial gyroscope	Smartphone (iPod, Apple, Cupertino, CA)	100 Hz	1	Back of the arm, 3 cm above upper edge of olecranon	Clinic	B-B score calculated from acceleration and angular velocity signals, ROM	Research

COM, center of mass; ROM, range of motion.

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Table 2. Study Characteristics—Athletic Training

Study	Country	Study Design	Sample Size	% Female	Age, y	Healthy	Injury/Surgery	Sport	Level
Fujishita et al., 2023[57]	Japan	Retrospective	39	51.3	13.7 [13-15]	Healthy		Multisport	Adolescent
Long et al., 2023[73]	U.S.A.	Descriptive observational	20	0	19.45 ± 1.69	Healthy		Runners	Collegiate
DeJong Lempke et al., 2022[70]	U.S.A.	Prospective cohort	21	61.9	20 ± 1	Healthy		Runners	Collegiate
Hamidi Rad et al., 2021[83]	Switzerland	Prospective observational	19	52.6	19.5 ± 2.7	Healthy		Swimming	Elite
DeJong Lempke et al., 2021[71]	U.S.A.	Prospective observational	64 (ERLLP, n = 32; healthy, n = 32)	75	21 ± 5 (ERLLP); 23 ± 6 (healthy)		ERLLP	Runners	Recreational
Caswell et al., 2020[76]	U.S.A.	Cohort study	49	100	16.2 ± 1.2	Healthy		Lacrosse	High school
Brooks et al., 2021[74]	Canada	Prospective observational	58	0	21.9 ± 1.7	Healthy		Football	Collegiate
Carey et al., 2021[75]	Australia	Observational case series	21	0	not recorded	Healthy		Rugby	Elite
Reyes et al., 2020[81]	Australia	Prospective observational	210	56.2	23.2 ± 3.6 (HAE above threshold), 24.5 ± 4.1 (HAE below threshold)	Healthy		Australian football	Professional
Caswell et al., 2019[77]	U.S.A.	Cross-sectional	77	0	16.6 ± 1.2	Healthy		Lacrosse	High school
Willmott et al., 2018[82]	Australia	Prospective cohort	53	35.8	25.97 ± 4.10 (M); 27.32 ± 4.40 (F)	Healthy		Australian football	Elite
King et al., 2018[80]	New Zealand	Prospective cohort	21	100	29.2 ± 7.8	Healthy		Rugby	Not recorded
Caswell et al., 2017[78]	U.S.A.	Descriptive epidemiology	35	100	16.2 ± 1.2	Healthy		Lacrosse	High school
Lauck et al., 2022[56]	U.S.A.	Prospective observational	41	0	Not recorded	Healthy		Football	Collegiate
Pexa et al., 2023[86]	U.S.A.	Prospective cohort	27	100	20.6 ± 1.4	Healthy		Soccer	Collegiate
Li et al., 2019[87]	U.S.A.	Case–control	115	0	26.1 ± 2.87	Healthy		Football	Professional
Benson et al., 2021[85]	Canada	Prospective observational	49	49	16.5 ± 0.6	Healthy		Basketball	High school
Khalil et al., 2021[66]	U.S.A.	Prospective observational	19	0	19.9 ± 1.2	Healthy		Baseball	Collegiate
Mehta, 2019[69]	U.S.A.	Observational	18	0	17.0 ± 0.7	Healthy		Baseball	High school
Freehill et al., 2023[84]	U.S.A.	Descriptive laboratory	11	0	[10-11]	Healthy		Baseball	Competitive

ERLLP, exercise-related lower leg pain; HAE, head acceleration event.

Performance Biomechanics and Training Optimization

Three studies equipped runners with shoe-mounted RunScribe Plus sensors set at a sampling frequency of 200 Hz.[70,71,73] Two of these evaluated running biomechanics in collegiate cross-country athletes.[70,73] Long et al.[73] explored changes in running mechanics during long, interval, and recovery run types during a one-week period, whereas DeJong Lempke et al.[70] collected data across an entire competitive season to compare these parameters with athletes' perceived exertion and self-reported wellness. In another study by DeJong Lempke et al.,[71] RunScribe sensors identified longer ground contact time as the primary differentiation between healthy runners and runners with exercise-related lower leg pain.

Hamidi Rad et al.[83] demonstrated the ability to evaluate swimming kinematic variables and performance using a single IMU attached to the sacrum with a waterproof band. The study team used a tethered speedometer and 2-dimensional cameras to validate swimmers' displacement/velocity and swim phase, respectively. In a laboratory study, Fujishita et al.[57] used a plantar pressure system and IMU to measure plantar pressure distribution and gait function in adolescent athletes with excessive rearfoot eversion. This system divided the sole into 9 distinct areas of plantar pressure according to 99 capacitive sensors.[57]

Workload Monitoring and Injury Prevention

Nine studies quantified head acceleration events (HAEs) during live contact sports (Australian football, American football, rugby, and lacrosse).[56,74-78,80-82] IMU-derived linear and rotational acceleration data were used to characterize magnitude, frequency, distribution, and type of load according to player position and mechanism of impact. Caswell et al.[76] also analyzed the effectiveness of implementing headgear in mitigating contact forces in high school girls' lacrosse, with the purpose of informing future optimization of equipment for improved game safety. The most common linear acceleration threshold was >10 g, ranging from >9.6 g to >15 g. Four studies limited analyses to impacts ≥20 g, and another also used a >30-g threshold for a case series analysis.[75-78,82] This removes lower acceleration events typically associated with routine game play activities. Reyes et al.[81] defined HAEs as >95.0 g for male and >85.5 g for female Australian football players and found that 5 players were screened for concussion after suffering verifiable in-game head impacts, despite having HAEs below threshold. This study team concluded that the X-patch alone does not reliably identify players for concussion screening, highlighting the need for video cross-referencing to verify these impacts. Furthermore, Willmott et al.[82] reported no association between X-patch measured HAEs and changes in post-game Sport Concussion Assessment Tool symptom severity.

Sensor-derived variables of physical workload during game-play were collected to examine associations with risk of injury. Khalil et al.[66] and Mehta[69] studied the relationship between Motus sleeve-derived throwing kinematics with anatomic adaptations of the medial elbow and development of upper-extremity injuries, respectively. Freehill et al.[84] demonstrated the feasibility and reliability of using a single IMU at the midhumerus to obtain pitch and throw counts for an alternative measure of baseball player load. Benson et al. evaluated workload in youth basketball players using the VERT, a validated IMU device that records jump count and jump height, while keeping surveillance of onset of tendinopathy and acute or overuse lower extremity injuries.[85] Pexa et al.[86] and Li et al.[87] evaluated predictors of delayed-onset muscle soreness in female collegiate soccer players and soft-tissue injury in professional American football players, respectively, using IMU-derived workload variables and trends in exertion throughout an entire season.

Rehabilitation

Sixteen studies applied wearable technology in rehabilitation programs and return to sport (RTS) assessments. Full study demographics are detailed in Table 3.[59,63,65,67,68,72,88-97]

Postinjury

Three studies evaluated patients' recovery during the acute period following mild traumatic brain injury (mTBI) or sports-related concussion (SRC).[88-90] Trbovich et al.[88] used the wrist-worn ActiGraph GT3x+ (Pensacola, FL) to correlate at-home patient sleep assessment with next-day SRC symptoms. This watch-like actigraphy device is a noninvasive option to objectively measure sleep and wake cycle parameters. Neely et al.[90] also evaluated post-mTBI sleep behavior, instead using a Fitbit containing photoplethysmography technology to record autonomic nervous (cardiovascular and respiratory) fluctuations. Fino et al.[89] demonstrated the diagnostic utility of using 4 IMUs attached to the forehead, lumbar spine, and shanks to quantify nonobservable, subtle motor deficits (head angular velocity and horizontal head turn) during clinical assessment of patients with SRC.

DeJong Lempke et al.[72] implemented a standard at-home exercise rehabilitation program in conjunction with outdoor sensor-based gait training with contact-time feedback using RunScribe Plus sensors to improve pain, function, and movement patterns in runners with exercise-related lower leg pain.

Table 3. Study Characteristics—Rehabilitation

Study	Country	Study Design	Sample Size	% Female	Age, y	Healthy	Injury/Surgery	Time of Follow-Up With Sensor (Postinjury/ Postsurgery)	Sport	Level
Lizzio et al., 2021[68]	U.S.A.	Controlled laboratory	20	0	17.8 ± 2.5	Healthy			Baseball	High school, collegiate
Leafblad et al., 2019[67]	U.S.A.	Descriptive laboratory	60	0	17.8 ± 2.2	Healthy			Baseball	Collegiate, high school
Dowling et al., 2018[65]	U.S.A.	Descriptive laboratory	95	0	16.5 ± 0.7	Healthy			Baseball	High school
DeJong Lempke et al., 2022[72]	U.S.A.	Randomised controlled trial	20	30	23 ± 4		ERLLP	>4 wk after onset of pain	Runners	Collegiate, recreational
Neely et al., 2022[90]	U.S.A.	Prospective observational	49	51	14.8 ± 1.8		mTBI	Postinjury, 2 wk following initial assessment	Multisport	High school
Trbovich et al., 2021[88]	U.S.A.	Prospective observational	17	47.1	15.35 ± 2.09		SRC	<72 h post-SRC, and throughout first month postinjury or until cleared for RTP	Multisport	Not recorded
Fino et al., 2019[89]	U.S.A.	Case–control longitudinal	49 (SRC, n = 24; healthy, n = 25)	25 (SRC); 24 (healthy)	20.3 ± 1.3 (SRC); 20.9 ± 1.4 (healthy)		SRC	Postconcussion, 9x over 8 wk	Multisport	Collegiate
Aufwerber et al., 2019[59]	Sweden	Case series	34	18	38.8 ± 8.7		ATR repair	2 wk and 6 wk postoperative	N/A	N/A
Moretti et al., 2023[96]	Italy	Case series	51	0	28.82 ± 5.33		ACLR	Postoperative, >1 y (20.75 ± 9.38 mo)	Soccer	Elite, semiprofessional
Dan et al., 2019[97]	Australia	Controlled laboratory	102 (ACLR, n = 65; normal, n = 27; elite athletes, n = 10)	28 (ACLR); 37 (normal); 100 (elite athletes)	33.84 ± 10.05 (ACLR); 25.93 ± 9.71 (normal); 22.80 ± 3.64 (elite athletes)		ACLR	8-15 mo post-op (ACLR group)	Multisport	not recorded
Mengis et al., 2023[63]	Switzerland	Prospective clinical trial	67	29.9	25.3 [21.9-32.0]		ACLR	Postsurgery (ACLR group), length of rehab	N/A	N/A
Bailey et al., 2016[94]	Canada	Case–control	51 (control, n = 26; ACLR, n = 25)	50 (control); 36 (ACLR)	21 ± 5 (control); 27 ± 10 (ACLR)		ACLR	5.6 mo postoperatively (SD: 2.6; range: 1.3–9.2)	N/A	N/A
Setuain et al., 2015[95]	Spain	Case control descriptive	22 (ACLR, n = 6; control, n = 16)	0	25.59 ± 1.01		ACLR	6.3 ± 3.4 y postoperative	Handball	Elite

(continued)

Table 3. Continued

Study	Country	Study Design	Sample Size	% Female	Age, y	Healthy	Injury/Surgery	Time of Follow-Up With Sensor (Postinjury/ Postsurgery)	Sport	Level
Bell et al., 2018[92]	U.S.A.	Cross-sectional	66 (ACLR, n = 33; control, n = 33)	66.7	20.3 ± 1.8 (ACLR); 20.8 ± 1.6 (control)		ACLR	27.8 ± 17.5 mo postoperative	N/A	N/A
Ezzat et al., 2021[93]	Canada	Cohort	102 (ACLR, n = 51; healthy, n = 51)	100	17.8 [14.6-22.6]		ACLR	1.1 (1.0-2.0) y postoperative	Multisport	not recorded
Gadangi et al., 2023[91]	U.S.A.	Prospective cohort	37	45.9	60.85 ± 7.94 (M); 57.41 ± 9.19 (F)		RCR	Preoperative and postoperative, 2-wk intervals	N/A	N/A

ACLR, anterior cruciate ligament reconstruction; ATR, Achilles tendon rupture; ERLLP, exercise-related lower leg pain; mTBI, mild traumatic brain injury; N/A, not available; RCR, rotator cuff repair; RTP, return to play; SD, standard deviation; SRC, sports-related concussion.

Postoperative Monitoring

Similar to Neely et al.,[90] Gadangi et al.[91] used triaxial accelerometry and polysomnography within the Whoop 2.0 wrist strap to examine pre- and postoperative sleep patterns and compare these metrics to PROMs after rotator cuff repair. This was the only included study to report on WSD patient compliance, which decreased from 84% at surgery to 46% at 6 weeks' postoperatively. Aufwerber et al.[59] used a plantar force sensor insole and pedometer to monitor mechanical loading and step count in patients treated with early functional mobilization in a dynamic orthosis after Achilles tendon rupture repair.

Seven studies used WSD data to monitor recovery and outcomes following ACL reconstruction (ACLR). Time from surgery to assessment with WSDs is recorded in Table 3. Mengis et al.[63] provided patients with a single IMU worn over the tibial tuberosity to monitor knee joint function throughout the length of their postoperative rehabilitation. This sensor was synchronized with the Orthelligent HOME smartphone application, which provided patients and clinicians instant feedback following exercise and test performance to track rehabilitation status and progress. Bell et al.[92] and Ezzat et al.[93] both used the ActiGraph GT3X, consisting of a triaxial accelerometer worn at the anterior superior iliac spine, to measure time spent in physical activity and moderate-to-vigorous physical activity at a mean of 2.3 and 1.1 years after ACLR, respectively. Bailey et al.[94] and Setuain et al.[95] evaluated acceleration profiles, jumping biomechanical deficits, and leg symmetry using a single IMU strapped over the L3 vertebra during a step-up-and-over test and vertical jump test battery, respectively, in patients undergoing ACLR. This anatomical location was chosen to best approximate the patient's center of mass. Moretti et al.[96] also collected IMU-derived biomechanical performance parameters, with the purpose of evaluating the role of concomitant meniscus repair in elite soccer players who returned to sport after ACLR. In a controlled laboratory study, Dan et al.[97] used accelerometry data to evaluate varus and valgus alignment differences for ACLR RTS assessment. These coronal alignment abnormalities are well-recognized risk factors for graft rupture and may not always be accounted for by routine functional RTS testing. A MatScan pressure sensing mat was included for secondary analysis of symmetry. No metrics were identified that indicated when the appropriate time to RTS should be recommended, although future analysis is promising.

Return to Throwing

Three laboratory-based studies used a Motus sleeve to evaluate throwing mechanics in healthy high school and collegiate baseball players during a typical long-toss return-to-throw program that is commonly

implemented in shoulder rehabilitation.[65,67,68] Similar in design, each used a single sleeve-embedded IMU sensor worn just distal to the medial epicondyle of the throwing arm humerus with a sampling rate of 1,000 Hz. Quantifying the variability of these throwing metrics at interval distances is used to guide individualized programs and progression during return-to-throwing rehabilitation. These principles can be used as a template for all RTS activities.

Research

Seven studies applied WSDs for novel research applications to optimize future clinical assessment, screen for injury risks, classify injury types, and estimate bone and tendon forces.[58,60-62,64,79,98] These study demographics are detailed in Table 4.[58,60-62,64,79,98] Four studies trained a machine learning algorithm using wearable signals as input features.[58,60,79,98] Sample sizes for the algorithms were variable and limited, ranging from 9 to 64 subjects.

Discussion

The current landscape of wearable devices in sports medicine is heterogeneous but rapidly evolving. The majority of selected studies within this review used IMU-derived data and reported diverse methodologies, technological interventions, and outcome metrics. To date, WSDs have been used in sports medicine within athletic training, rehabilitation monitoring, and future research applications. Wearable technology is increasingly used by clinicians, sports scientists, physical therapists, athletics trainers, teams, and individual athletes to identify injury risk, aid in diagnosis, and optimize training safety, management, and outcomes. WSDs offer an accessible and reliable method for collecting patient data both actively and passively—from physical function (step count, acceleration, sleep, heart rate) to PROMs related to the athlete's mental health and joint function. However, mass adoption within the clinical aspect of sports medicine is yet to longitudinally manifest from the preoperative injury state to postoperative athletic return to prior performance.

According to a recent survey, 20.7% of multidisciplinary trauma teams currently leverage wearable sensors, with WSD use limited to simple activity outcomes such as step count.[99] Despite limited use in clinical practice, there are various applications of WSDs in arthroplasty and trauma literature. Van der Walt et al.[100] reported significantly greater step count in patients using an accelerometer activity tracker after hip and knee arthroplasty. Edwards et al.[101] found that although PROMs and pain outcomes normalized earlier in the postoperative period following total shoulder arthroplasty, accelerometers demonstrated a delayed motion recovery in comparison, highlighting the need for further insight into functional outcomes. Using pressure insoles, Braun et al.[102] concluded that patients recovering from lower-extremity fractures significantly deviated from standard weight-bearing recommendations—underscoring this newfound ability to measure patient compliance and accommodate more pragmatic rehabilitative goals. Ramkumar et al.[103] conducted a longitudinal study using 2 IMUs for monitoring of knee flexion during daily exercise sessions up to 12 weeks post-total knee arthroplasty. This remote-patient monitoring system was able to holistically capture patient recovery via continuous data, mobility and joint-specific metrics, pain outcomes, and compliance within a central shared dashboard between patient and surgeon.

As previously reported, equipping athletes with wearable sensors enables real-time monitoring of workload across a variety of indoor and outdoor sport settings by collecting metrics that would not have been achievable in a restricted clinical or laboratory environment. Training program monitoring with this technology allows team trainers and medical personnel to develop personalized programs to optimize performance and reduce risk of overuse injury.[104-106] These portable sensors are also optimal for objective, within-clinic testing. For example, rather than using force plates for traditional ACLR RTS assessment via the step-up-and-over test, acceleration profiles from a single IMU can be a less expensive and more practical replacement.[94]

The future of WSDs in sports medicine holds exciting possibilities, with recent research emerging for advancements in precision monitoring, diagnostics, and personalized interventions. For example, Pichonnaz et al.[64] automatically measured the B-B score to evaluate shoulder function using a smartphone with a programmed application attached to the arm. Automatic scores were responsive and demonstrated excellent diagnostic power, suggesting its eventual use in routine clinic workflows for joint pathology evaluation. Di Paolo et al.[61] reported lower-limb joint kinematic differences between laboratory and field settings using an inertial sensor suit. This may have implications in redesigning future ACL injury–prevention strategies, which have largely been informed by laboratory-based risk factor assessments. Brund et al.[62] used inertial data to develop predictive equations to estimate patellar and Achilles tendon forces during running. This new tool can aid future epidemiological researchers to quantify cumulative loads and injury etiology with improved validity. The intersection of WSDs and artificial intelligence may hold considerable promise for diagnoses and prediction of likelihood of injury or recovery. Using linear and rotational accelerometry data and video verification, Goodin et al.[79] classified head and body impacts in football players with 94.67% true positive and 95.65%

Table 4. Study Characteristics—Research

Study	Country	Study Design	Sample Size	% Female	Age, y	Healthy	Injury/Surgery	Time of Assessment/ Follow-up (Injured Cohort)	Sport	Level	Purpose
McGeown et al., 2022[98]	New Zealand	Cross-sectional	17	35.3	21.0 [16-28]		mTBI	Postinjury, 17 days [13-18.5]	Multisport	Not recorded	Evaluate a convolutional neural network model trained with sensor data collected during the Buffalo Concussion Treadmill Test to classify patient subgroups
Braun et al., 2023[60]	Germany	Proof-of-concept	38	36.8	46.6 [21-67]		Musculoskeletal trauma	Preinjury and <6 wk postinjury	N/A	N/A	Use machine learning to predict functional recovery based on readily available pre- and postinjury activity
Elstub et al., 2023[58]	U.S.A.	Cross-sectional	9	33.3	Not recorded	Healthy			Runners	Recreational	Train a machine learning algorithm to estimate tibial bone forces
Goodin et al., 2021[79]	Australia	Prospective observational	64	6.3	26.3 ± 3.79	Healthy			Australian football	Elite	To automatically classify head and body impacts using an instrumented mouthguard, video-verified impacts, and machine-learning algorithms
Brund et al., 2021[62]	Denmark	Prospective observational	24	29.2	26 ± 1.3	Healthy			Runners	Recreational	To develop predictive algorithms to estimate patellar and Achilles tendon forces and impulses per stride during running
Di Paolo et al., 2023[61]	Netherlands	Cross-sectional	28	100	14.9 ± 0.9	Healthy			Soccer	Adolescent	To investigate kinematic differences between laboratory and on-field settings

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Table 4. Continued

Study	Country	Study Design	Sample Size	% Female	Age, y	Healthy	Injury/Surgery	Time of Assessment/ Follow-up (Injured Cohort)	Sport	Level	Purpose
Pichonnaz et al., 2015[64]	Switzerland	Prospective cohort	108	50 (rotator cuff); 78 (fracture); 60 (capsulitis); 43 (instability); 50 (control)	63.5 ± 10.6 (rotator cuff); 60.1 ± 15.6 (fracture); 52.5 ± 13.8 (capsulitis); 32.1 ± 14.1 (instability); 28.2 ± 6.2 (control)		Shoulder pathology (rotator cuff, fracture, capsulitis, instability)	<2 wk following medical consultation (or 6 wk post-stabilization), and again 6mo later	N/A	N/A	To determine the measurement properties of the shoulder function B-B score measured with a smartphone

mTBI, mild traumatic brain injury; NA, not available.

true negative detection in test sets. McGeown et al.[98] used ECG and accelerometry data to classify predominant symptom cluster (PSC) subgroups in patients with mTBI, with moderate classification accuracy (92% for physiological PSC and 60% for vestibulo-ocular PSC). Braun et al.[60] used pre- and postinjury wearable step count data to predict a patient's functional recovery status after 6 weeks, with predictive accuracy of more than 80%. Using pressure insoles and IMU signals, Elstub et al.[58] estimated peak tibial bone force in runners with an average error of 5.7% compared with calculations using motion capture and a force instrumented treadmill.

Although evident that wearable technology has powerful potential benefits for monitoring and management, there are many challenges to be addressed before effective and accurate widespread adoption is assured. Although imminently achievable, current applications of wearable technology remain fragmented across various vendors, with no central platform catering to the longitudinal patient experience pathway of the injured athlete's RTS and daily living. Much of the literature to date has primarily assessed the validity and reliability of an array of devices and lacks consistency across usage. Furthermore, evaluations of inertial sensors are largely performed in a controlled laboratory environment, using small sample sizes, and employing a variety of testing protocols, outcome measures, and body locations. Future research should focus on inter-study consistency and reaching standardization of sensor usage guidelines and outcome metrics for generalizability of findings. High patient compliance is necessary to facilitate the integration of WSDs in routine clinic workflows and at-home monitoring. Poor data collection may be due to patient demographic factors and the feasibility of device use in daily settings or over long-term. Braun et al.[60] demonstrated a novel approach to wearable data collection in orthopaedic clinics which may be less partial to patient cooperation, compliance, and recall bias.[60] Rather than employing a dedicated WSD, often postinjury, the study team obtained data from smartphones or wrist-worn devices already used by patients before injury. Although affected by age-related distributions of technology usage and limited to basic activity assessment, this "bring your own device" strategy allows for post-treatment metrics to be compared to baseline, pre-injury activity data for personalized recovery.[107] Older age has been associated with the inability or disinterest in using WSDs to track activity.[108] This may be due to usability concerns, such as accessibility, personalization, and interface design. Due to the inherent nature of WSD data gathering, data security and ethical concerns regarding the collection, use, and dissemination of personal information may be additional accessibility barriers.[109]

Limitations

This scoping review has limitations. Due to challenges presented by the broad range of sensor types and study designs within this review, a quality assessment was not performed. The technology remains new and thus applications and device limitations are inherent to the novelty of these wearables. Validation studies were excluded at the discretion of the reviewing author, and only available full-text articles were included. A substantial number of included studies evaluated HAEs during high-contact gameplay. However current commercial sensors (i.e., X-patch and GForce Tracker) can lead to considerable measurement bias when detecting HAEs based on a linear acceleration threshold.[110,111] For on-field deployment, video verification remains crucial to provide contextual information of HAEs and prevent false positives.[112]

Conclusions

WSDs enable real-time monitoring of human performance across a variety of implementations and settings, allowing collection of metrics otherwise not achievable. WSDs are powerful tools with multiple applications within athletic training, patient rehabilitation, and orthopaedic and sports medicine research.

Disclosure

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